

Second Year Research Proposal

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Executive Summary: I will use the HETDEX spectroscopic survey, along with Bagpipes, a Bayesian SED fitting tool, to uncover how galaxy properties and Ly α strength are correlated using a sample of roughly 1000 galaxies in the North Ecliptic Pole. This sample will be used to improve estimates of the emergent Ly α emission from distant galaxies which will improve constraints on reionization.

Introduction

An important period in the history of the universe is the epoch when the universe transitioned from a neutral phase to an ionized phase. Evidence from quasars point to the universe completing this phase transition by $z \approx 6$ but the precise time of its onset, its speed and its underlying driving mechanism are still unsolved questions [1]. Uncovering this information will give us crucial insight about the universe's first ionizing sources, but progress in this field has been impeded by the limited number of useful tracers. One promising tracer that can shed light on reionization is the Lyman Alpha (Ly α) recombination line. Ly α is observed to be bright in star-forming galaxies and such galaxies are abundant in the early universe (e.g., [2]; [5]), implying that Ly α may be able to trace reionization. Because Ly α resonantly scatters with the intervening neutral hydrogen, our ability to detect Ly α is regulated by processes within the galaxy, such as dust content, dust kinematics and dust distribution, and the IGM, where Ly α can trace the neutral IGM fraction at a given epoch (e.g., [12], [10], [4] and references therein). The emergent Ly α provides details about the physical properties of the galaxy that regulate its escape, and also the state of the IGM it traversed near where it was emitted.

Using Ly α as a tracer assumes that we know beforehand the intrinsic Ly α flux of a galaxy, which to date has been a limiting factor in previous studies (e.g., [13],[11],[8]). What is needed is to tie the emergent Ly α emission to the physical properties of an individual galaxy, such that we can use the observed properties of individual galaxies to predict the emergent Ly α flux. Previous works were limited by *small* sample sizes and *biased* sample selection. For example, spectroscopic follow-up typically requires pre-selection which results in a bias towards the most luminous sources, complicated further by everyone using different selection criteria. To generate a comprehensive analysis on how galaxy properties correlate with emergent Ly α flux what is needed is to have a *larger*, and *unbiased* sample of galaxies *which I will be working on for my second year project*.

Second year Project

I intend to leverage the Hobby-Eberly Telescope Dark Energy Experiment (HETDEX) spectroscopic survey to generate a sample of Lyman alpha emitters (LAEs) and explore their Lyman alpha strength based on their physical properties. The HETDEX survey is an **un-biased, spectroscopic survey** looking at 450 deg^2 with 300 deg^2 in the northern sky and 150 deg^2 in the equatorial sky, to uncover LAEs between $1.9 < z < 3.5$ [6]. I will be working on the 10 deg^2 *Euclid* North Ecliptic Pole (NEP) deep field, which is being observed as

an extension to the main HETDEX fields to take advantage of the wealth of deep multi-wavelength imaging data already available in this field. The HETDEX instrument gathers spectra using a fiber integral field unit (IFU) called the Visible Integral-Field Replicable Unit Spectrograph (VIRUS) [7]. VIRUS’s wavelength coverage ranges from 3500-5500 Å allowing for optimal Ly α detection from galaxies at $1.9 < z < 3.5$. HETDEX has in place an automated pipeline that:

- 1) reduces the data
- 2) finds emission line(s) in the spectrum and
- 3) attempts to classify the line it detects using Bayesian techniques [9].

These features make HETDEX the ideal survey to use for finding a large sample of LAEs for correlation analysis.

To utilize the most of what HETDEX has to offer we use the Hawaii-Two-0 (H20) imaging and photometric survey of the NEP field. The H20 team is currently reducing data in the NEP field where they have deep imaging and are getting photometry using a software called the Farmer, which models each of the sources and uses the best model between point sources, extended source or a combination to force the model to match up to the image and get the correct flux from the source. The imaging will be used in the Lyman alpha classification as that method uses imaging information, such as size and magnitude to come up with a probability of the galaxy being an LAE. While the photometry will be used by Bagpipes to get galaxy properties.

Project Analysis Plan

Fall 2021

Currently, the Hawaii-Two-0 (H20) survey is $> 75\%$ complete, with the remaining needed nights expected to be allocated in 2021 and as part of our sharing agreement, we are provided with fully reduced imaging and photometric catalogs. The data that I have on hand is one HSC photometric catalog in one pointing in HETDEX. I also have access to the HDR2.1 curated catalog. With more data coming in as the fall semester progresses. We expect to have a near full photometric catalog of the NEP field from the H20 team once they are done re-reducing their data and more HETDEX data with the release of HDR3 in the fall.

During the fall of 2021, I will be working on regions where VIRUS and H20 have reached their full depths and are fully reduced. The current region I am working on can be seen in the blue box at the top left of Figure 2. Using the HETDEX curated catalog with proper LAE cuts, ie the probability of the line emission being Lyman alpha, $\chi^2 < 1.2$ and signal to noise (SN) > 5 , show that in this small region there is about 100 sources. With the addition of more photometric data from the H20 team and HETDEX’s HDR3 data release we expect the sample size to increase to 10x this amount.

Careful consideration was put to ensure that we are extracting on Lyman alpha emitters. To help with this we are targeting galaxies whose photometric redshift falls between $1.9 < z < 3.5$. By implementing this photometric redshift cut we reduce the amount of low- z interlopers showing up in our analysis. However, this method is only good if we trust the photometry and after some tests we decided that the HSC photometry is very robust. This is best shown in Figure 1 where it shows comparison between their phot- z and spec- z where with the added IRAC imaging, it is able recover the correct redshift.

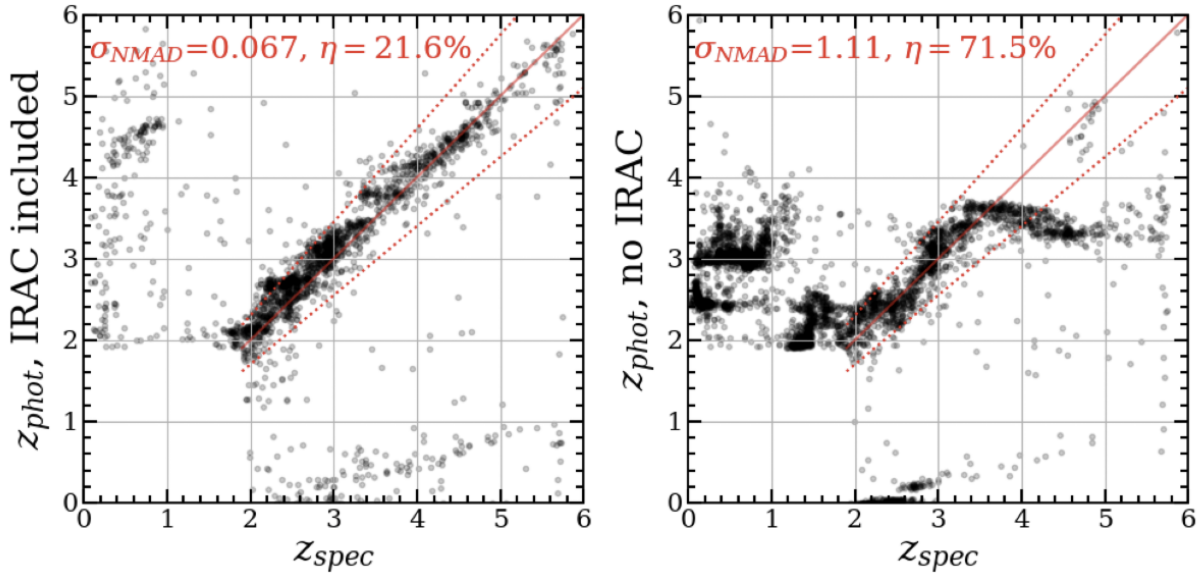


Figure 1: The HSC photometric redshift comparing it to known spec-z show that HSC photometric redshift does a good job at getting the right redshifts of galaxies. Even more so in the range of $1.9 < z < 3.5$, this plot also shows how useful having IRAC imaging is to constraining the photometric redshift

Once I get my sample of high- z sources I need to get their spectrum and find Lyman alpha emission. At first we tried extracting on the HSC photometric coordinates but found that the signal of the lines were all really low. Even comparing it with known matches of the same source to the HETDEX curated catalog. So what I will work on for the first half of the fall of 2021 is to implement a grid search approach that is centered on the HSC coordinate to find where the peak SN of the Lyman alpha line is coming from. Once I have found where the line SN is maximized and at what wavelength that occurs, I then extract on that position using Elixer to generate an Elixer report. Once the Elixer report has been generated I then impose another set of cuts for the quality of the line and line emission. These cuts include:

- 1) $SN > 5$
- 2) $P_{lya} > .7$
- 3) $\chi^2 < 1.2$

These added cuts ensure that the line measured by Elixer is a strong detection, and that there is a high confidence that the line is Lyman alpha. We do this latter test in case the photometric redshifts were wrong in the pre-selection process and we can get rid of them moving forward.

During the later half of the fall of 2021 I will be increasing the sample size as HDR3 will be out. I will spend some time getting familiar with the data and how to access what I need to perform my analysis. Once all that is done I will be at the stage where I can fit these Lyman alpha emitters with a Bayesian spectral energy distribution (SED) fitting code called Bagpipes. I will do this by using Subaru imaging and photometry which reaches 5σ depths in the following bands: $g = 27.1$, $r = 26.7$, $i = 26.4$, $z = 26$, $y = 24.9$ and *Spitzer*/IRAC

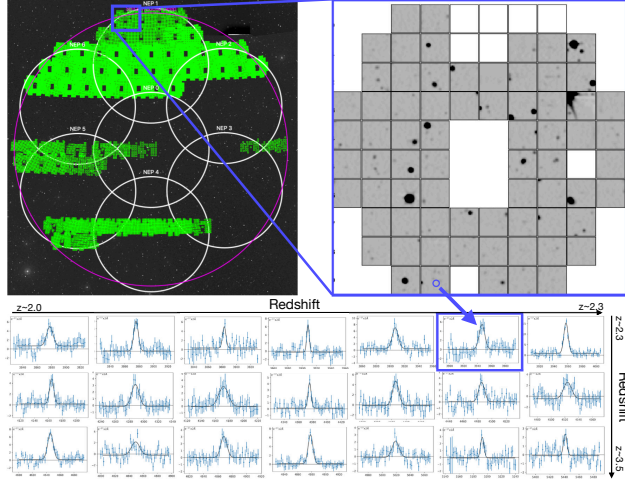


Figure 2: Imaging footprint of the NEP field. My current data set is comes from the top of the image. As of the writing this proposal the current data available with HETDEX is outlined by the yellow squares, where the outer circle outlines the NEP field layout, the inner white circles are the HSC-H20 pointings. We expect to increase our sample 10 fold once we get more photometric data from HSC and with the release of HDR3 in mid-fall.

$3.6\mu m$, $4.5\mu m = 25$, and use **Bagpipes** to determine the physical properties of the galaxies in my sample [3]. Bagpipes uses a built in stellar library to fit photometric points and does so by maximizing the likelihood that the spectrum comes from a set of parameters. Every time bagpipes fits the photometry with a set of properties we get that information in the form of a posterior distribution. One of the advantages of using the HETDEX survey is that we are able to have spectroscopic redshift for the galaxies in my sample, which eliminates one source of uncertainty by *not* incorporating a photometric redshift when performing the SED fits, as photometric redshifts are very much model dependant. Thus for these fits I will be fixing the redshift at the spectroscopic redshift when performing the fits.

Spring 2022

During the spring of 2022, I will be running bagpipes on galaxies I still need to fit to get galaxy properties but the spring will primarily consist on working on correlating galaxy properties to Lyman alpha emission. With the fitted galaxies from Bagpipes, I will now have a pretty big sample of galaxy properties to explore which will allow me to explore a wider dynamic range in Lyman alpha emission that previously done before. The goal is to generate a massive table of galaxy properties and bin them on the basis of galaxy property and Lyman alpha emission. I will explore correlation between properties such as; galaxy mass, dust attenuation, star formation rate (SFR) and specific SFR (sSFR) to see which global property contributes to the escape of Lyman alpha. The method I wil use is the Spearman correlation because it is sensitive to outliers but does assume that the errors are normally distributed which for most of the posteriors are close to normal. I will also explore any redshift evolution by also binning my galaxies according to redshift and seeing if I see an evolution with a galaxy property and Lyman alpha emission. The goal is to make diagnostic plots and a table that we can provide the community to reference that for a given set of galaxy properties, there is a distribution of Lyman alpha strength behind that. This will help with reionization work by tying it back to an individual galaxy's physical properties on a per galaxy basis.

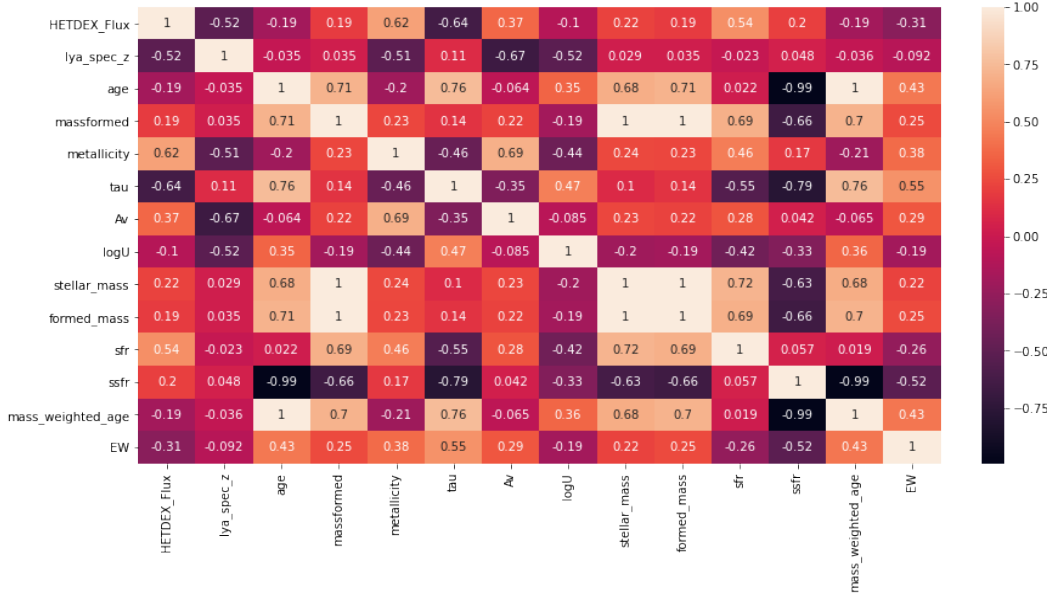


Figure 3: A diagnostic plot that I will make once my sample has been finalized and ran through bagpipes. This is a heat map showing how correlated the different parameters are to each other but the columns/rows that will be of interest to me is the first and last row/column in this heatmap that show how each galaxy property is to Lyman alpha strength.

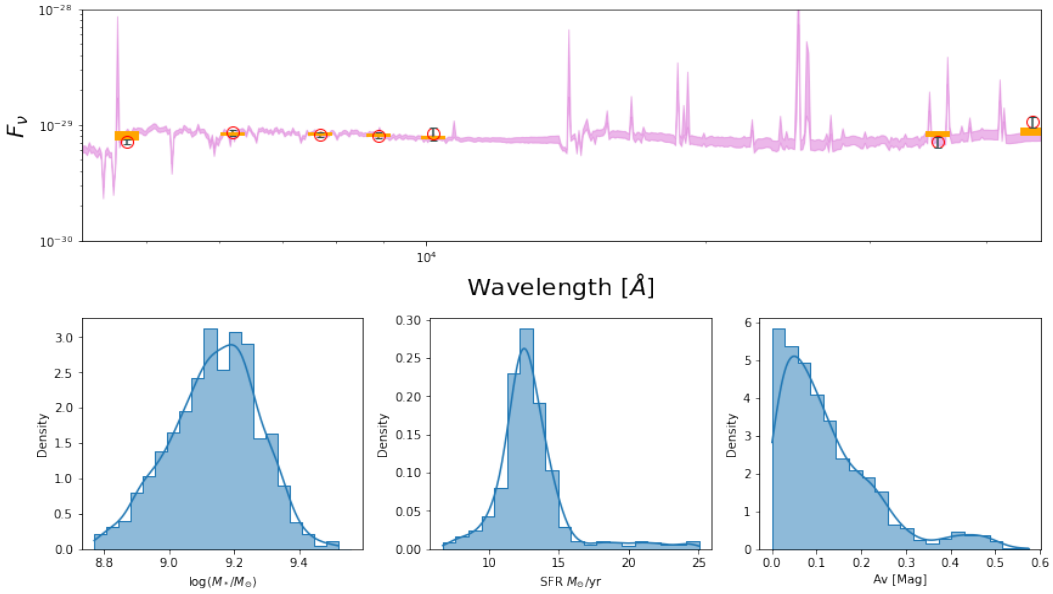


Figure 4: A plot demonstrating the output from bagpipes. The top panel shows one sigma range of the fits to the photometric data points. The red circles are the HSC and *Spitzer*/IRAC photometry and the blue is bagpipes 16th-84th percentile range. The bottom panel shows Bayesian posterior distributions of a couple of the galactic properties that I will be studying in this data set, namely the star formation rate, dust attenuation and galaxy mass.

References and Acknowledgement

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